

SOLE SOURCE JUSTIFICATION

Below the Simplified Acquisition Threshold (\$15,000 - \$350,000)

Project Title: ChemiDoc Go Imaging System with Image Lab Touch Software

A. Name and address of the proposed contractor(s)

Bio-RAD
2000 Alfred Nobel Drive,
Hercules, CA 94547

B. Describe why only the requested supply/service will meet the government's needs

Market research was conducted to identify commercially available gel and blot imaging systems capable of meeting National Institutes of Health laboratory requirements. The ChemiDoc Go incorporates a 20.4-megapixel back-illuminated complementary metal oxide semiconductor detector, providing high pixel density and signal-to-noise performance necessary for:

- Small-format gel imaging
- Accurate band densitometry
- Stain-free normalization workflows
- Continuity with previously generated datasets

Alternative systems utilize lower-resolution charge-coupled device sensors or line-scan architectures that do not match current imaging parameters used in validated laboratory standard operating procedures. The ChemiDoc Go includes a built-in 9.7-inch touchscreen color interface, enabling full standalone operation without the need for an external computer.

This capability is essential to:

- Avoid National Institutes of Health information technology cybersecurity configuration requirements
- Eliminate additional workstation procurement
- Reduce validation burden
- Simplify training and workflow integration

Some competing systems (for example, laser line-scan systems) require external personal computer control, introducing additional information technology and security overhead.

C. Describe why only the proposed contractor is capable of providing the requested supply/service

Based on the market research conducted and the documented technical requirements, it has been determined that the Bio-Rad ChemiDoc Go Imaging System is the only product capable of meeting the National Institutes of Health laboratory's minimum needs. Sole-source procurement is therefore appropriate. Bio-Rad Laboratories is the only responsible source capable of providing the ChemiDoc Go Imaging System with the required combination of detector specifications, integrated standalone operation, validated Smart Tray compatibility, and footprint requirements without unacceptable delay or programmatic risk.

Failure to procure the Bio-Rad ChemiDoc Go would result in:

- Mandatory revalidation of imaging assays
- Delays in National Institutes of Health-supported research studies
- Increased costs associated with workflow transition
- Potential disruption of longitudinal data continuity
- Additional information technology compliance burden for personal computer-dependent systems

These impacts would adversely affect program timelines and resource utilization. Bio-Rad provides integrated service, consumables, and imaging accessories calibrated for the ChemiDoc Go platform. Maintaining a unified vendor ecosystem ensures reproducibility and long-term operational stability.